AI ASSISTED CODING

ASSIGNMENT-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

• Prompt AI to convert a Python class with attributes and methods into equivalent Java code.

• Verify that constructors, method signatures, and data types are properly handled.

• Run both versions (Python & Java) to compare expected outputs.

Expected Output:

• Equivalent working Java class translated from Python.

```python
# create a python code using class with attributes and methods
class BookCollection:
    def __init__(self):
        self.books = []

    def add_book(self, book: str) -> None:
        self.books.append(book)

    def remove_book(self, book: str) -> None:
        if book in self.books:
            self.books.remove(book)
        else:
            print(f"Book '{book}' not found in the collection.")

    def list_books(self) -> list:
        return self.books
# Example usage:
if __name__ == "__main__":
    bc = BookCollection()
    bc.add_book("To Kill a Mockingbird")
    bc.add_book("1984")
    print(bc.list_books())  # Output: ['To Kill a Mockingbird', '1984']
    bc.remove_book("1984")
    print(bc.list_books())  # Output: ['To Kill a Mockingbird']
    bc.remove_book("The Great Gatsby")  # Output: Book 'The Great Gatsby' not found in the collection.
```

```java
# convert the above code to Java language
import java.util.ArrayList;
import java.util.List;
public class BookCollection {
    private List<String> books;

    public BookCollection() {
        this.books = new ArrayList<>();
    }

    public void addBook(String book) {
        books.add(book);
    }

    public void removeBook(String book) {
        if (books.contains(book)) {
            books.remove(book);
        } else {
            System.out.println("Book '" + book + "' not found in the collection.");
        }
    }
    public List<String> listBooks() {
        return books;
    }

    public static void main(String[] args) {
        BookCollection bc = new BookCollection();
        bc.addBook("To Kill a Mockingbird");
        bc.addBook("1984");
        System.out.println(bc.listBooks()); // Output: [To Kill a Mockingbird, 1984]

        bc.removeBook("1984");
        System.out.println(bc.listBooks()); // Output: [To Kill a Mockingbird]

        bc.removeBook("The Great Gatsby"); // Output: Book 'The Great Gatsby' not found in the collection.
    }
}
```

```
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/lab_assignment-19-1.py
['To Kill a Mockingbird', '1984']
['To Kill a Mockingbird']
Book 'The Great Gatsby' not found in the collection.
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> 
```

```
Output                                                    Clear
[To Kill a Mockingbird, 1984]
[To Kill a Mockingbird]
Book 'The Great Gatsby' not found in the collection.

=== Code Execution Successful ===
```

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

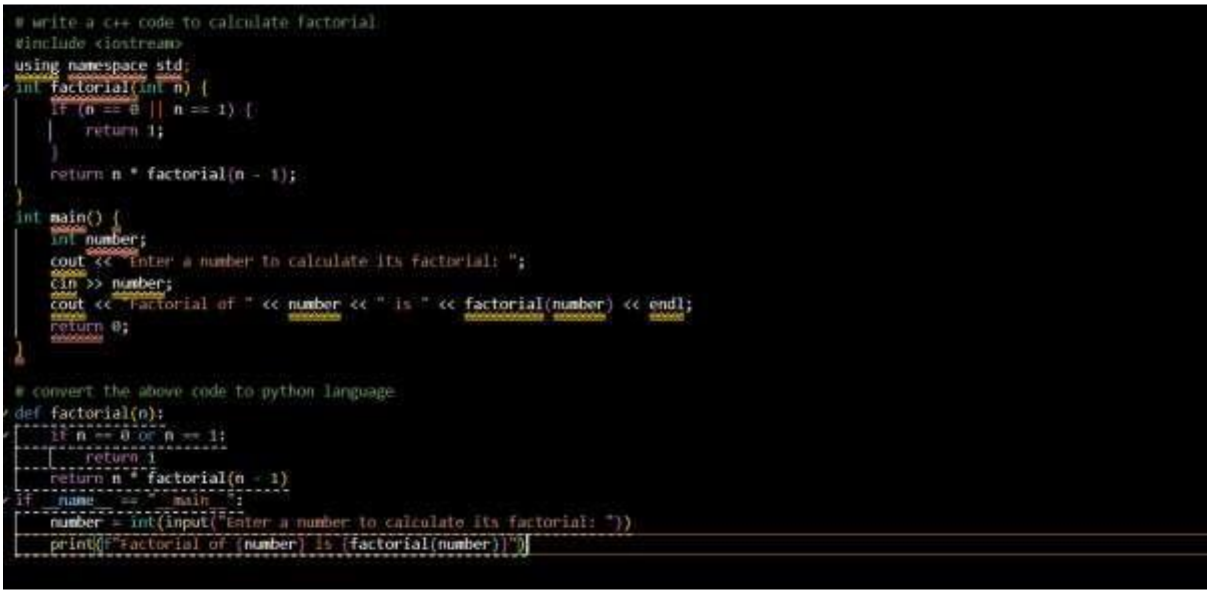
• Ask AI to translate a recursive C++ factorial function into Python.

• Validate correctness by testing with sample inputs.

• Ensure Python version follows Pythonic conventions.

Expected Output:

• A correct Python implementation equivalent to the original C++ factorial function.



Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

• Provide AI with a SQL query (e.g., SELECT name, salary

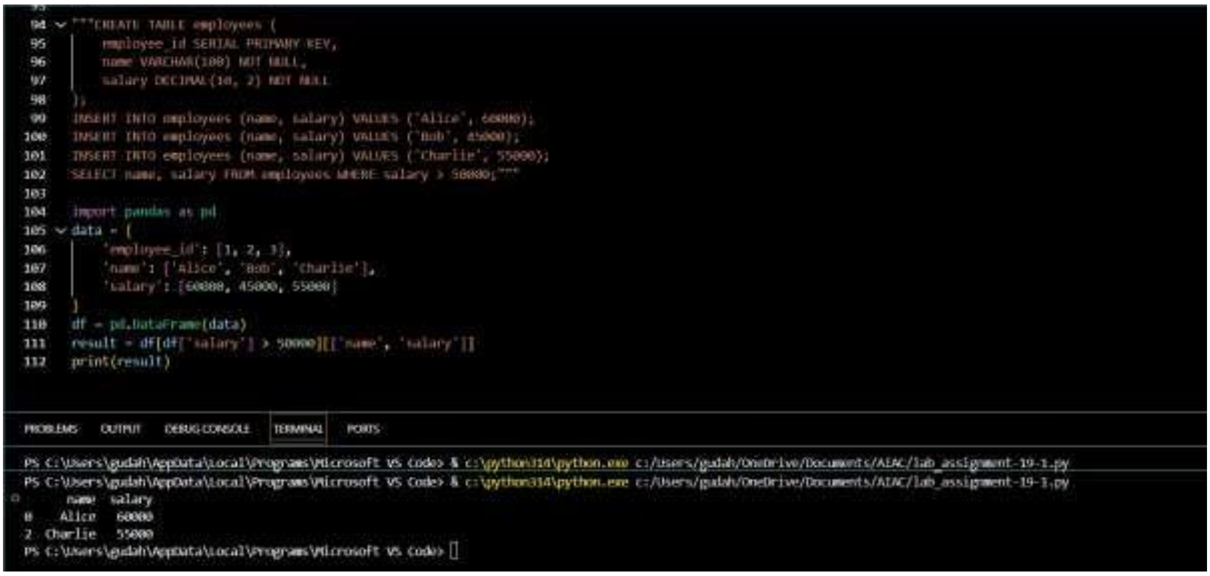
FROM employees WHERE salary > 50000). •

Ask AI to generate equivalent Pandas code.

• Test the generated code on a sample DataFrame.

Expected Output:

• Equivalent Pandas query that retrieves the correct subset of data.



Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

• Converting a Python web scraping snippet into JavaScript for browser automation.

• Translating a Java method into C# for enterprise applications.

Instructions:

• Prompt AI to translate code between two chosen languages.

• Test both implementations to ensure they produce the same results.

• Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

• Working equivalent code snippets in at least two different programming languages.

```
#generate a python code that print N natural numbers
from py_compile import main
N = int(input("Enter a number: "))
for i in range(1, N+1):
    print(i)

#Conver the above code to  C language
#include <stdio.h>

int main() {
    int N;
    printf("Enter a number: ");
    scanf("%d", &N);
    for (int i = 1; i <= N; i++) {
        printf("%d\n", i);
    }
    return 0;
}

#convert the above code to Go
package main
import "fmt"
func main() {
    var N int
    fmt.Print("Enter a number: ")
    fmt.Scan(&N)
    for i := 1; i <= N; i++ {
        fmt.Println(i)
    }
}
```

```
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/code_translation.py
Enter a number: 5
1
2
3
4
5
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code>
```

Challenges: Differences in libraries, syntax (indentation vs braces), and execution models (sequential vs asynchronous).

Resolution: Used async/await in JavaScript, adjusted output formats, and verified functional equivalence.

Learning: Translation requires adapting to language paradigms and selecting equivalent libraries.